

Study Guide for Exam #1 with Practice Problems

MATH 240 — Introduction to Statistics, Fall 2026

General Information

The first exam will be held in-class on Friday, October 2nd. Keep in mind the following things:

- You will have 75 minutes to complete the exam, unless you have an accomodation.
- This is a **closed-book** exam, with the exception of a single 8.5x11" sheet of paper of notes. You can use both sides of the sheet to write down anything.
- You will not need to write any R code on this exam. However, you must able to interpret R output (e.g. summary tables, plots, metrics, etc.).
- Cell-phones, laptops, calculators, and any other electronic devices are not permitted. No interaction with anyone is allowed.

Expect the exam to be divided up into three parts:

- Part I: Multiple choice
- Parts II and III: Situational or based on a dataset

There will be about 5-10 multiple choice questions, following by 3-4 questions for each of the remaining parts. All of the topics below are fair game for these questions.

Topics

- Ch. 1-3: Introduction to Data
 - Data basics (types of variables, relationships, observational studies vs. experiments)
 - Study design (populations vs. samples, parameters vs. statistics, biases, sampling methods, experimental design)
- Ch. 4-6: Exploratory Data Analysis
 - One categorical variable (bar plots, summary tables)
 - One numerical variable (histograms, bar plots, five-number summaries)

- Two variables (contingency tables, faceted/stacked/side-by-side plots, scatter plots)
- Analysis of each type of plot (histograms, bar plots, box plots, scatter plots)
- Ch. 7: Simple Linear Regression
 - Getting least-squares estimates (and their interpretations)
 - Use of correlation and R^2 values
 - Fitting a model and writing its least-squares line
 - Diagnosing a model fit
 - Interpretations of estimates, determining significance
 - Prediction of y values with least-squares line

Practice Problems

1. Ch. 1: Exercises 2, 3, 7, 12, 14, 19
2. Ch. 2: Exercises 2, 7, 11, 14, 17, 25
3. Ch. 4: Exercises 2, 3, 4, 5, 7
4. Ch. 5: Exercises 2, 5, 11, 26
5. Ch. 7: Exercises 1, 5, 21, 23